

Week 38 exercises – INF201 2026

Task 1 – Clean up sensor readings

You have been asked to clean up a set of sensor readings. The readings are given as a list where each element is either a floating-point number or None (NoneType, indicating a temporary sensor failure). You are also given a floating-point number to use as a threshold for removing low readings. A sensor reading is considered valid if the value is equal to or greater than this threshold. A reading of None is also not considered valid.

- Write a function that filters out non-valid readings and returns the cleaned-up list
- Write another function that reports the results of this clean up as a string with the format indicated in week38_task1.py.

For this assignment, you may assume that the list of readings is reasonably short, with no more than 100 elements. Assume sensor values to be between 0 and 20.000 (that is twenty, not twenty thousand).

Remember to document your functions and use type hints.

The file week38_task1.py contains an example list of sensor readings, threshold and sensor name. In your script, include the results you get using these as input.

Task 2 – Working with lists

The file week38_task2_broken.py contains three functions to do with creating a board for a game and evaluating game scores. Unfortunately, each function has an error. Your task is to find and correct these errors. Please include comments in each function, stating why the original code was wrong.

If you are struggling to understand what the code does, you could try visualizing it using [Python Tutor](#).

Hand-in information

- Remember to include filename extension for all submitted files
- You may submit Python scripts (.py) or Jupyter notebooks (.ipynb)
- If you submit a Jupyter notebook, please upload a pdf version as well
- AI chat-log can be submitted in its entirety or summarized (see lecture 02)